Class #13 - The Gram-Schmidt Process

1. Theorem: If $\{u_1, \cdots, u_p\}$ is an orthonormal basis for a subspace W of $\mathbf{R^n}$, then

$$\text{Proj}_W y = _____.$$

If $U = [u_1\ u_2\ \cdots\ u_p]$, then

$$\text{Proj}_W y = _____ \forall y \in \mathbf{R^n}.$$

Proof:

2. The Gram-Schmidt process is an algorithm for producting an _____or _____basis for any _____subspace of $\mathbf{R^n}$.

3. Example: Let $b_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$, $b_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \\ 1 \end{bmatrix}$, $b_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix}$.

Then $\{b_1, b_2, b_3\}$ is a basis for a subspace W of _____. Construct an orthogonal basis $\{u_1, u_2, u_3\}$ for W.

(Step 1) Let $u_1 = b_1$ and $W_1 = \text{Span}\{b_1\} = \text{Span}$ _____.

(Step 2) Let $u_2 = b_2 - \text{Proj}_{W_1} b_2$. (Then u_2 is orthogonal with _____).

(can scale u_2 to simplify hand calculations)

(Step 3) Let $W_2 =$ Span _____.

Then $\text{Proj}_{W_2} b_3 =$ _____.

Let $u_3 = b_3 - \text{Proj}_{W_2} b_3$. Then u_3 is orthogonal with _____.

Hence, $W =$Span$\{u_1, u_2, u_3\}$ where $\{u_1, u_2, u_3\}$ is an _____basis.
And an orthonormal basis for W is _____.

4. The Gram-Schmidt Process:
Given a basis $\{b_1, \cdots, b_p\}$ for a subspace W of $\mathbf{R^n}$, define

$$u_1 = _____$$

$$u_2 = _____$$

$$u_3 = _____$$

$$\vdots$$

$$u_p = _____$$

Then $\{u_1, \cdots, u_p\}$ is an _____basis for W.

5. Fill in the blank: Any nonzero subspace W of $\mathbf{R^n}$ has an _____basis.

6. The QR factorization:
If A is an $m \times n$ matrix with linearly independent columns, then A can be factored as

$$A = QR$$

where Q is an $m \times n$ matrix whose columns form an orthonormal basis for _____, and R is an _____upper triangular _____matrix with _____entries on its diagonal.

Proof: The columns of A form a basis for _____.
Construct an orthonormal basis $\{u_1, u_2, \cdots, _____\}$ for $W = \text{Col } A$.
Let $Q = [u_1 \ u_2 \ \cdots \ u_n]$.

For $i = 1, \cdots, n$, $a_i \in$Span$\{u_1, u_2, \cdots, u_i\}$. So there are constants such that

$$a_i = r_{1i}u_1 + \cdots + r_{ii}u_i + 0u_{i+1} + \cdots + 0u_n.$$

Can we always choose $r_{ii} \geq 0$?

Let

$$r_i = \begin{bmatrix} r_{1i} \\ \vdots \\ r_{ii} \\ 0 \\ \vdots \\ 0 \end{bmatrix}.$$

That is, $a_i =$ _____ for $i = 1, \cdots, n$.

Let $R = [r_1 \ r_2 \ \cdots \ r_n]$.

Then, $A = [a_1 \ a_2 \ \cdots \ a_n] = [Qr_1 \ Qr_2 \ \cdots \ Qr_n] = QR$.

Why is R invertible?

So, R has _____entries on its diagonal.